

Part 1 of 2

Host Software for RFID Based Attendance Management System



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In this article we describe how to design the user interface (UI) for a hardware and link data received from the hardware to a database on Windows PC. An RFID reader module and tags are used as hardware, and Visual C# (or C sharp) as the software-programming tool, to create a UI for the attendance management system. The software takes data from an RFID reader and other user-related data and writes/reads/updates it in the database. To limit the scope, this article does not deal with details of the hardware.

Designing the UI program

Aim of the software. This software has a UI with two tabs, Registration and Employee Data, as shown in Fig. 1. On clicking Registration tab, the admin inputs user-related data and registers (saves) it in the database. On clicking Employee Data tab, the admin can view a particular employee's data in the UI by entering the employee's ID in a UI field. The system architecture is shown in Fig. 2.

Pre-requisites. It requires Microsoft Visual Studio 2010, Microsoft SQL Server 2008, SQLEXPRESS_x64_ENU and SQLManagementStudio_x64_ENU.exe as software tools. We also need a readymade RFID reader module, capable of scanning RFID tags, and thereby sending read data to a host system using RS232 or USB (virtual UART), and a backend data-

base server running locally on your PC. Your PC should be connected to the RFID reader by means of RS232 or USB (virtual UART) and running a client UI for direct interaction with the user. Prior knowledge of C#

language would help in quick understanding of the topics discussed here.

Software architecture. The software architecture block diagram is shown in Fig. 3. Host software consists of client-server architecture with a service layer sandwiched between the client and server layers. This is done with the intention to create an abstraction between the client and server in order to reduce interdependency between both layers.

Along with client-server architecture, this software follows a multi-layered architectural approach. The entire program is divided into following parts:

1. **Presentation layer (client-side).** This is written in Windows presentation framework (WPF) using C#, which provides a rich UI to the user.

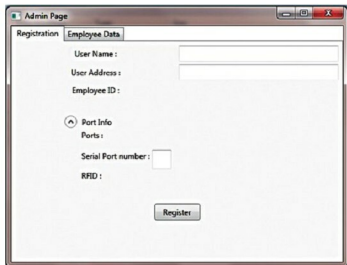


Fig. 1: Main UI program output

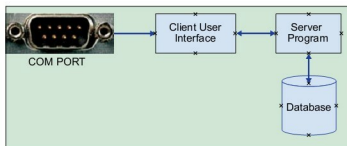


Fig. 2: System architecture

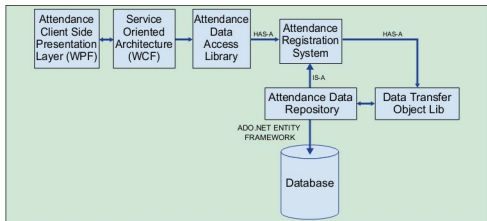


Fig. 3: Block diagram of the software architecture